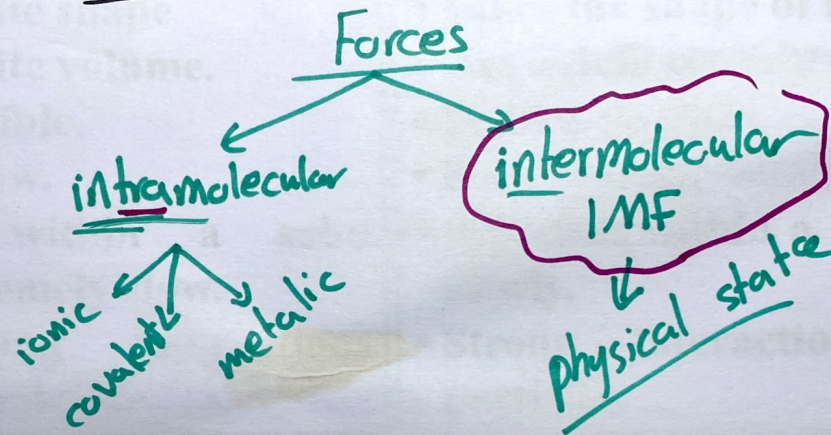


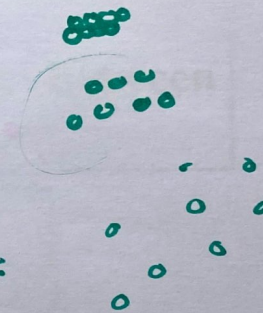
تَشَارِكْ

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
Ch. 11
Liquids and intermolecular Forces



gas < Liquid < solid

solid $\xrightarrow[\text{dec } P]{\text{inc temp}}$ Liquid $\xrightarrow[\text{dec } P]{\text{inc temp}}$ gas



Some Characteristic Properties of the States of Matter

Solid State

- Has a definite shape
- Has a definite volume.
- Incompressible.
- Does not flow.
- Diffusion within a solid occurs extremely slow.
- Very strong interactions between particles.

Liquid State

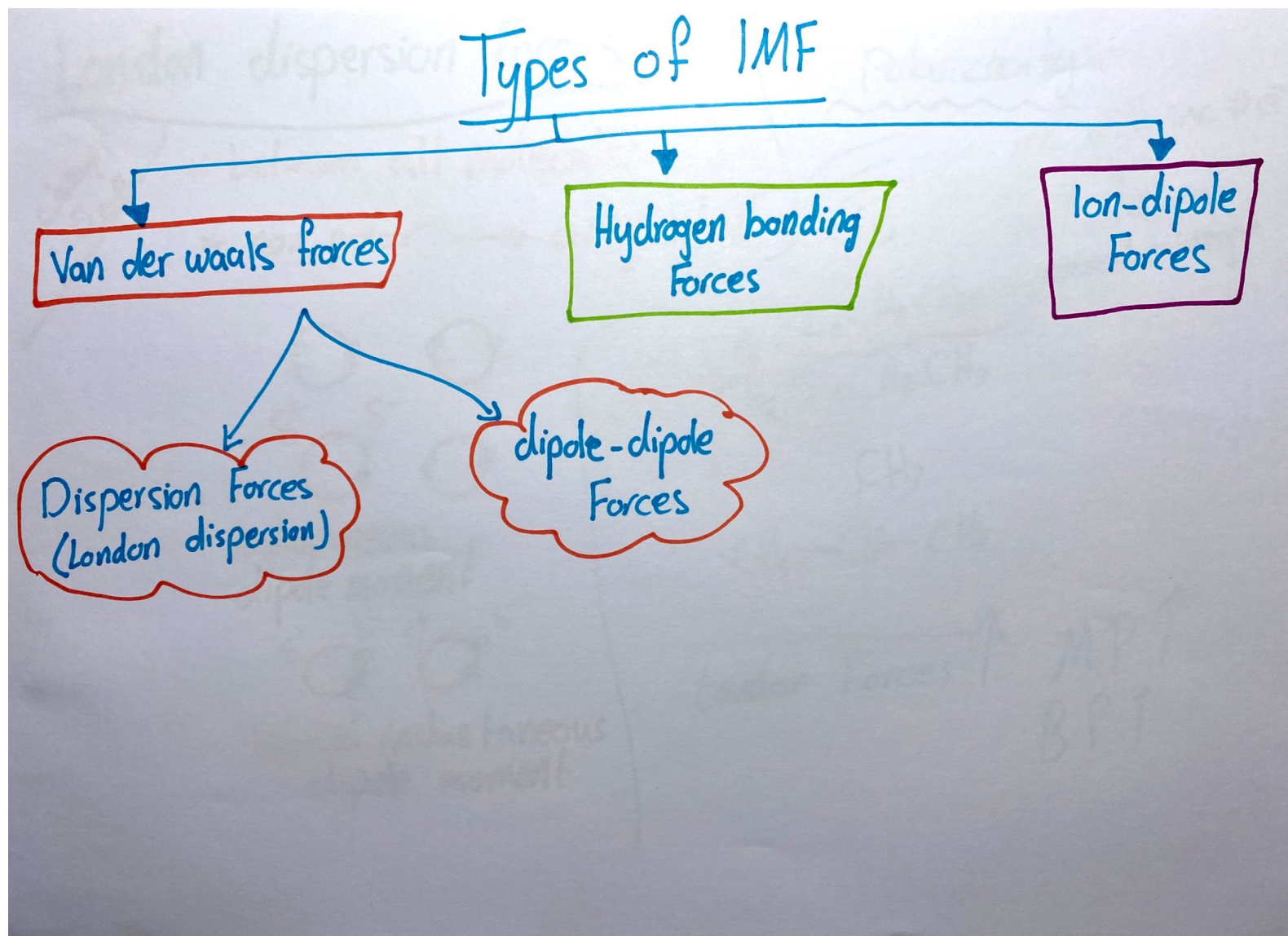
- Takes the shape of its container.
- Has a definite volume
- Incompressible.
- Flow readily.
- Diffusion within a liquid occurs slowly.
- Strong interactions between particles.

Gas State

- Takes the shape of its container.
- Takes the volume of its container.
- Compressible.
- Flow readily.
- Diffusion within a gas occurs rapidly.
- Negligible interactions between particles.



Dr. Ziyad A. Taha

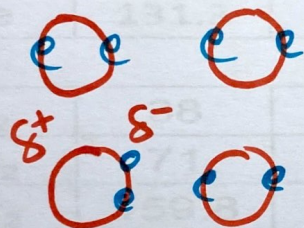


London dispersion Forces

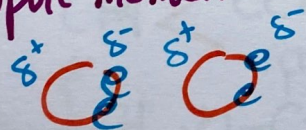
القوة
الضعيفة

* between all molecules.

* non polar $\rightarrow$ only $\rightarrow$ L.F



instantaneous dipole moment



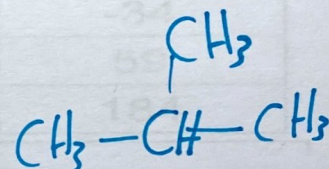
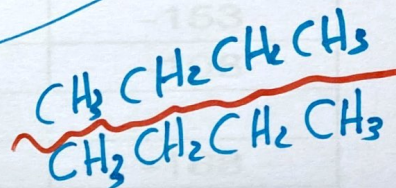
induced instantaneous dipole moment

Polarizability



inc with inc #e⁻

inc with inc molecular weight

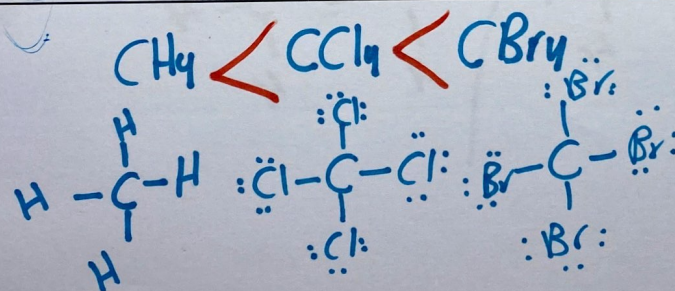


London Forces $\uparrow$

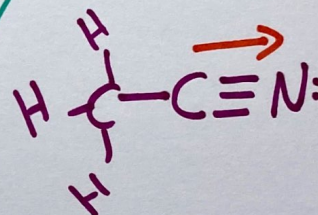
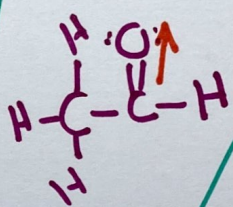
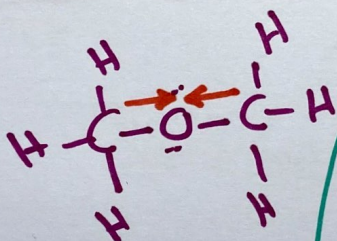
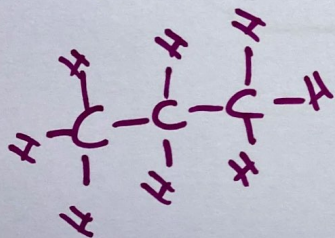
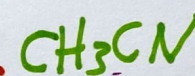
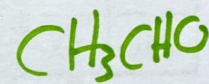
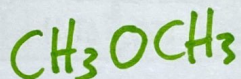
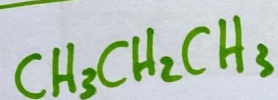
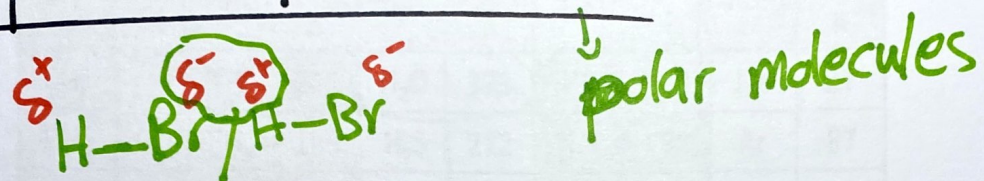
M.P $\uparrow$
B.P $\uparrow$

	M. wt	Melting Pt. (°C)	Boiling Pt. (°C)
He	4	-270	-269
Ne	20	-249	-246
Ar	39.9	-189	-186
Kr	83.8	-157	-153
Xe	131.3	-112	-108
F ₂	38	-220	-188
Cl ₂	71	-101	-34
Br ₂	159.8	-7	59
I ₂	153.8	114	184

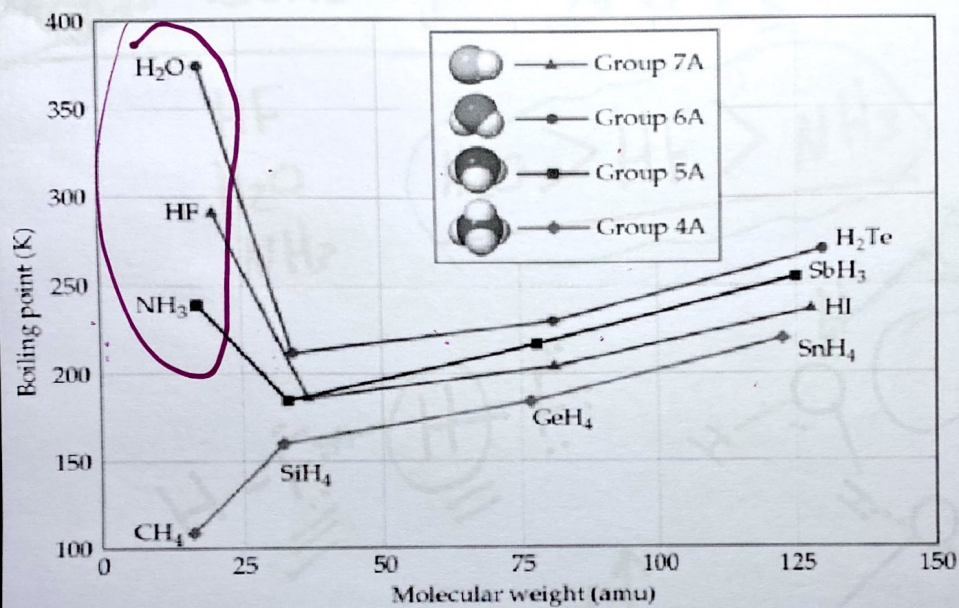
M.W ↑
 London ↑
 B.P ↑
 M.P ↑



Dipole-Dipole Forces $\uparrow$ M.P $\uparrow$ B.P $\uparrow$



XH_4	Bpt /K	XH_3	Bpt/ K	H_2X	Bpt/ K	HX	Bpt/ K	atom	Bpt/ K
CH_4	112	NH_3	240	H_2O	373	HF	293	Ne	27
SiH_4	161	PH_3	185	H_2S	212	HCl	188	Ar	87
GeH_4	184	AsH_3	218	H_2Se	232	HBr	206	Kr	121
SnH_4	221	SbH_3	256	H_2Te	271	HI	238	Xe	166



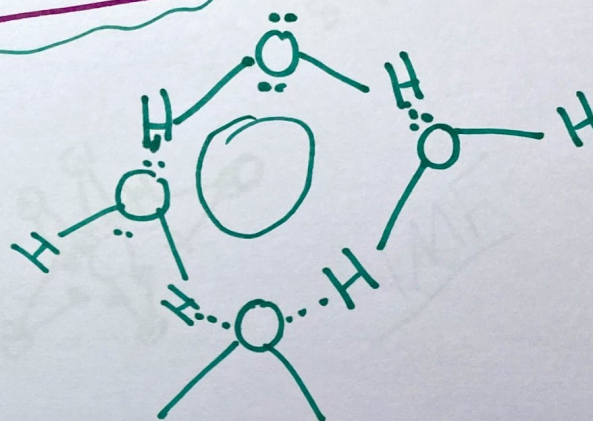
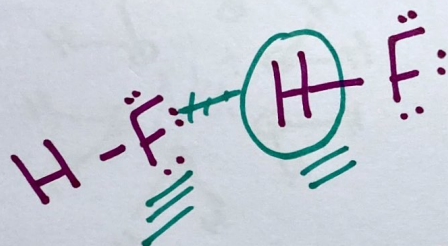
H-bond

Hydrogen - Bonding

NOF

HF
H₂O
NH₃

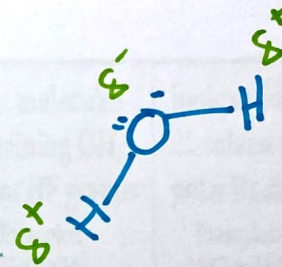
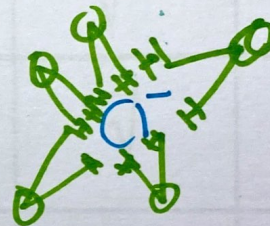
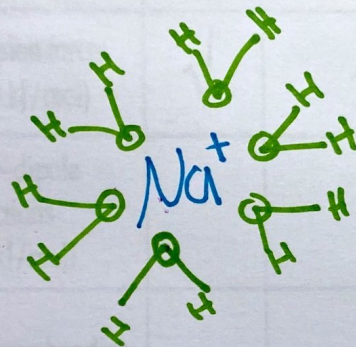
H₂O > HF > NH₃



H-bond ↑
E ↑
Size ↑

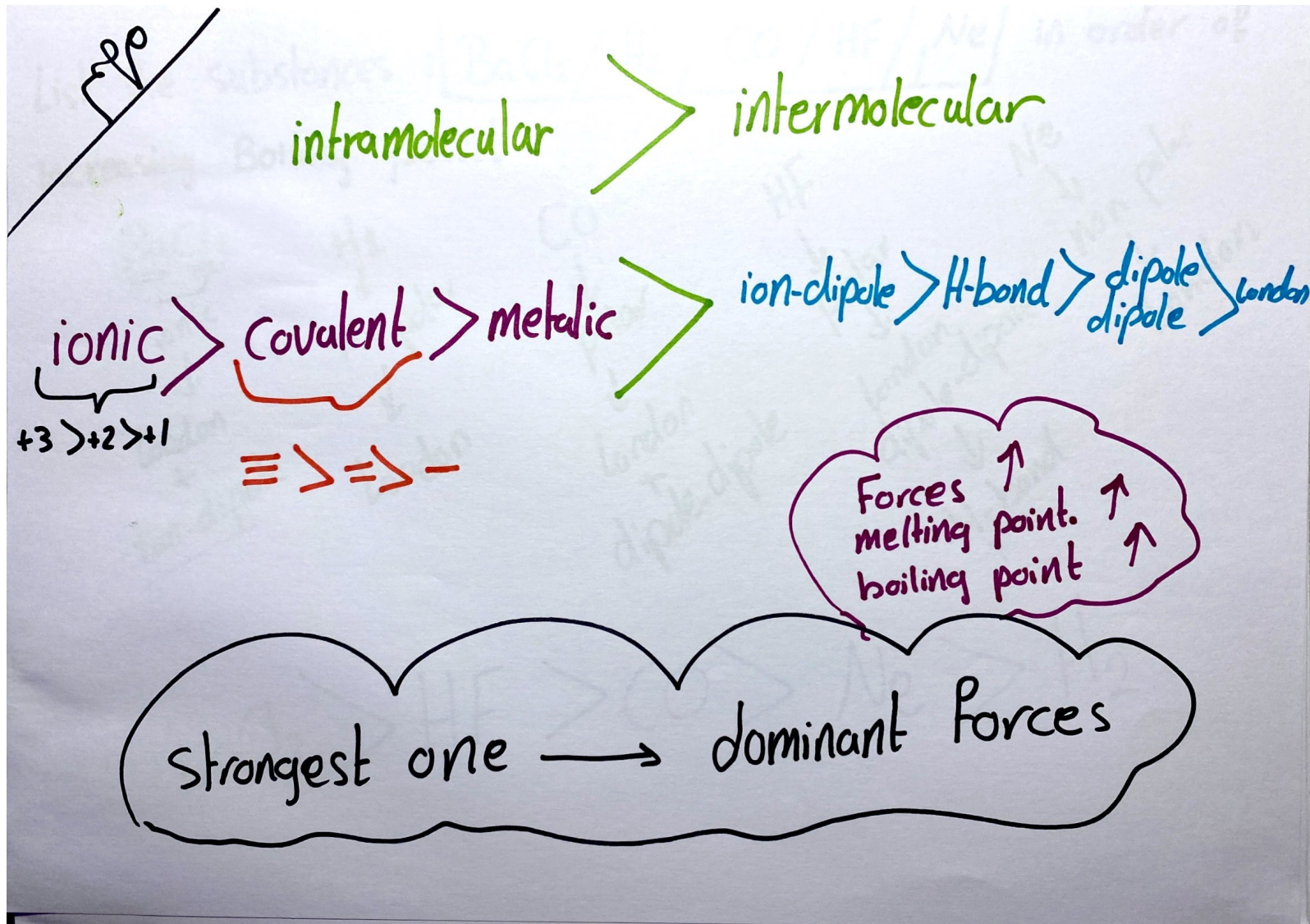
HF > H₂O > NH₃

Ion dipole forces



IMF

Type of intermolecular interaction	Atoms Examples: Ne, Ar	Nonpolar molecules Examples: BF_3 , CH_4	Polar molecules without OH, NH, or HF groups Examples: HCl, CH_3CN	Polar molecules containing OH, NH, or HF groups Examples: H_2O , NH_3	Ionic solids dissolved in polar liquids Examples: NaCl in H_2O
Dispersion forces (0.1–30 kJ/mol)	✓	✓	✓	✓	✓
Dipole–dipole interactions (2–15 kJ/mol)			✓	✓	
Hydrogen bonding (10–40 kJ/mol)				✓	
Ion–dipole interactions (>50 kJ/mol)					✓



List the substances : BaCl₂ / H₂ / CO / HF / Ne in order of increasing Boiling point.



ionic

↓
London

+
ion-dipole



↓
non polar

↓
London



↓
polar

↓
London

+
dipole-dipole



↓
polar

↓
London

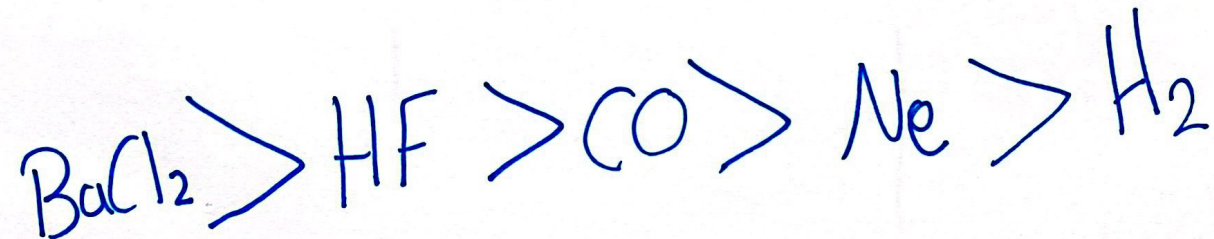
dipole-dipole

↓
H-bond



↓
non polar

↓
London



What types of IMF exist between each of the following molecules? determine the dominant one.

NOF ^H

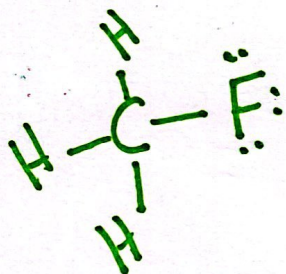
HBr
London
✓ dipole-dipole

CH₄
✓ London

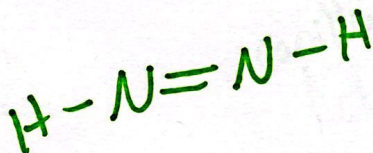
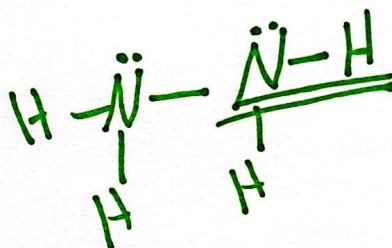
HF
London
dipole-dipole
✓ H-bond.



London
✓ dipole-dipole



London
dipole-dipole
✓ H-bond

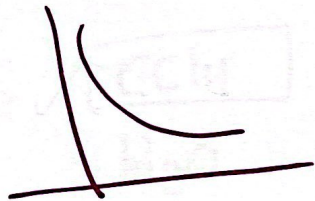


London
✓ ion-dipole

Properties of Liquids

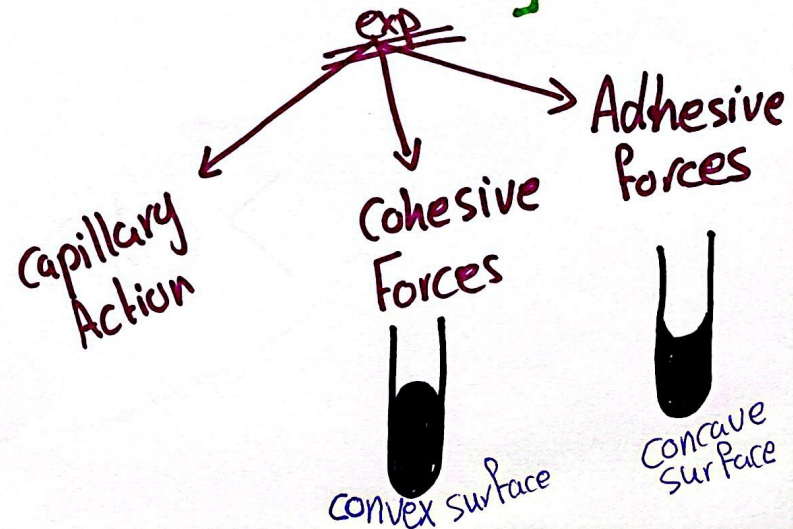
Viscosity

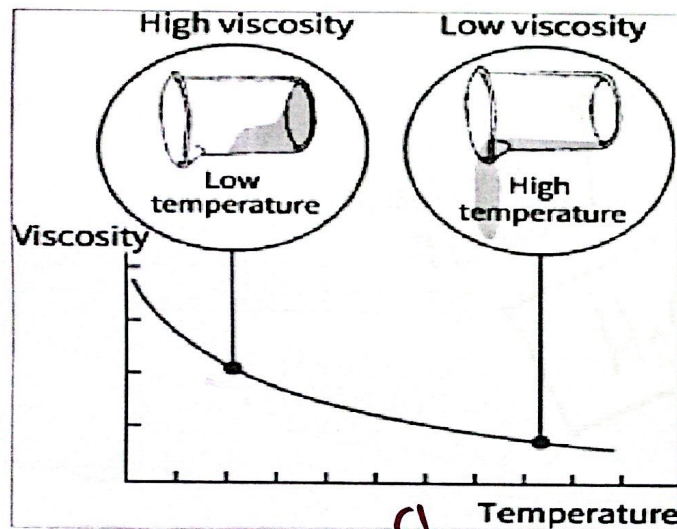
[resistance of a liquid to flow]



Surface Tension

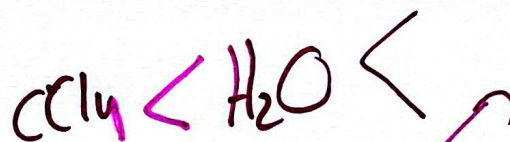
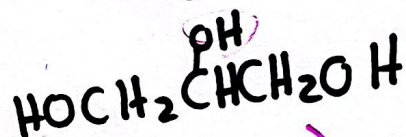
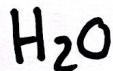
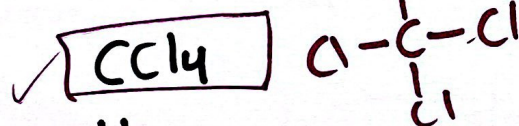
[results from imbalance of IMF at the surface of the liquid]



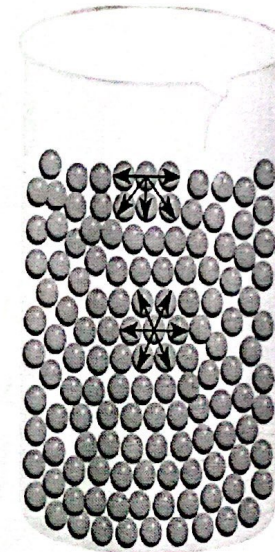


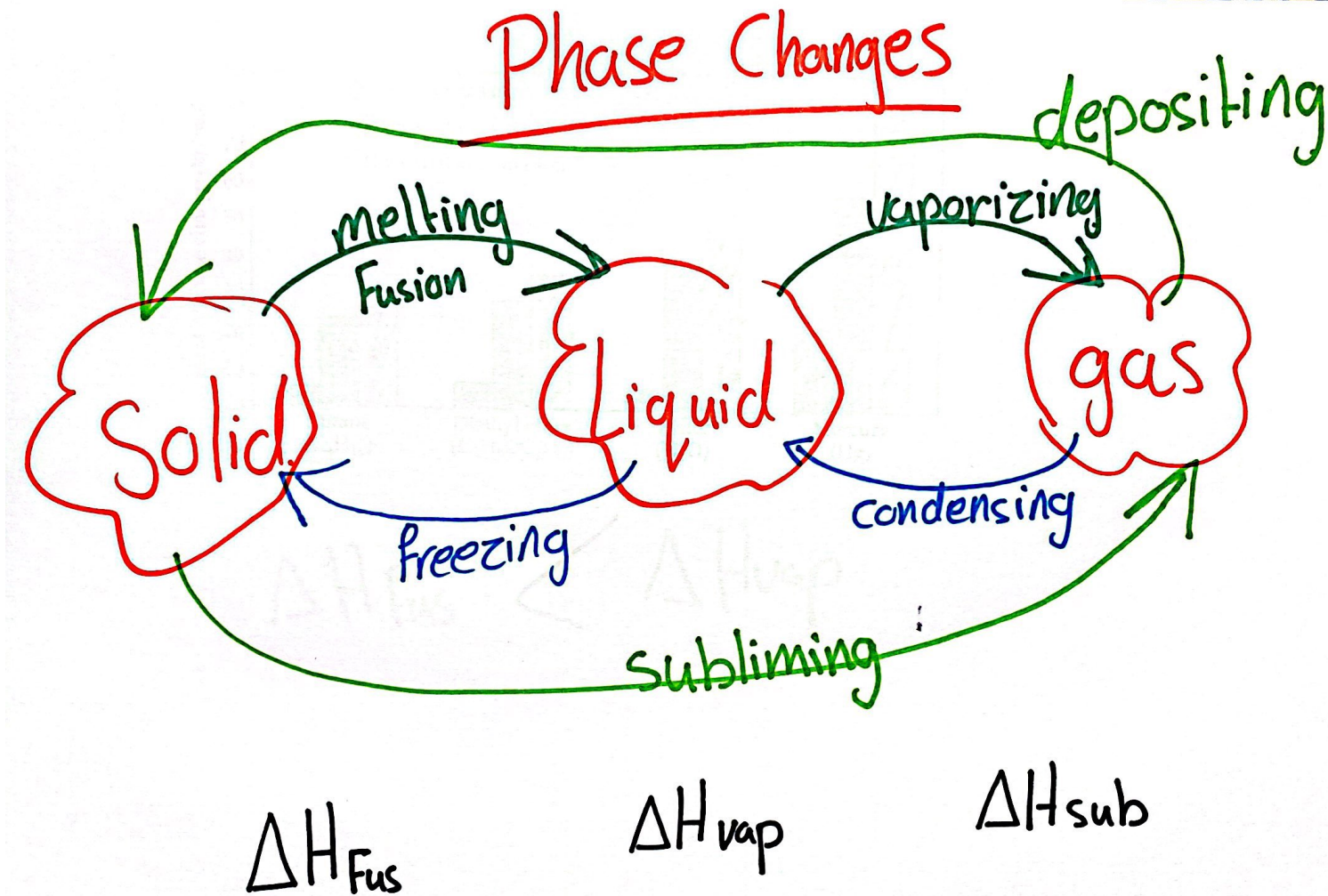
Viscosity $\uparrow$
 Molecular size $\uparrow$
 IMF $\uparrow$

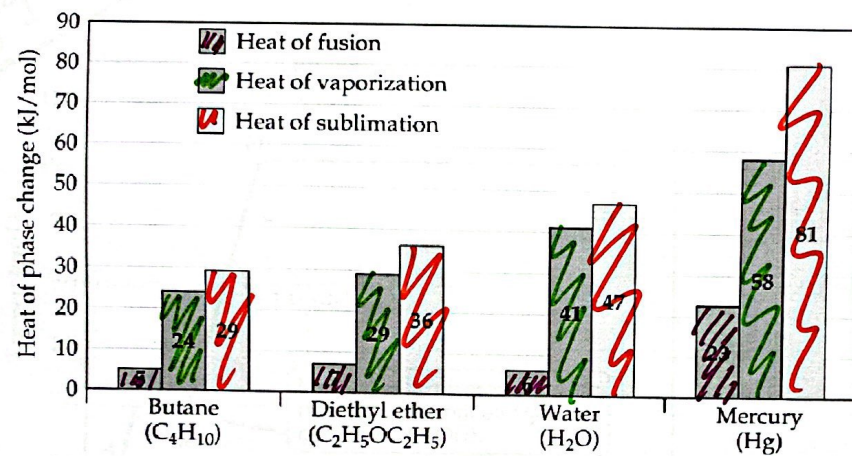
Viscosity $\downarrow$
 temp $\uparrow$



Surface Tension ↑ IMF ↑
H₂O highest

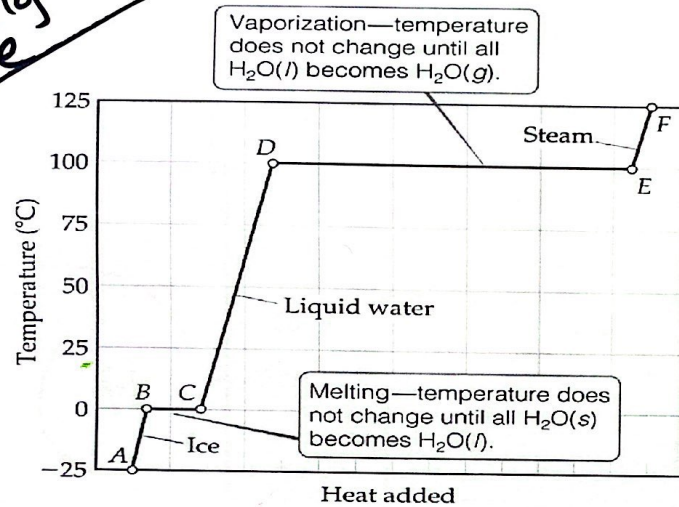




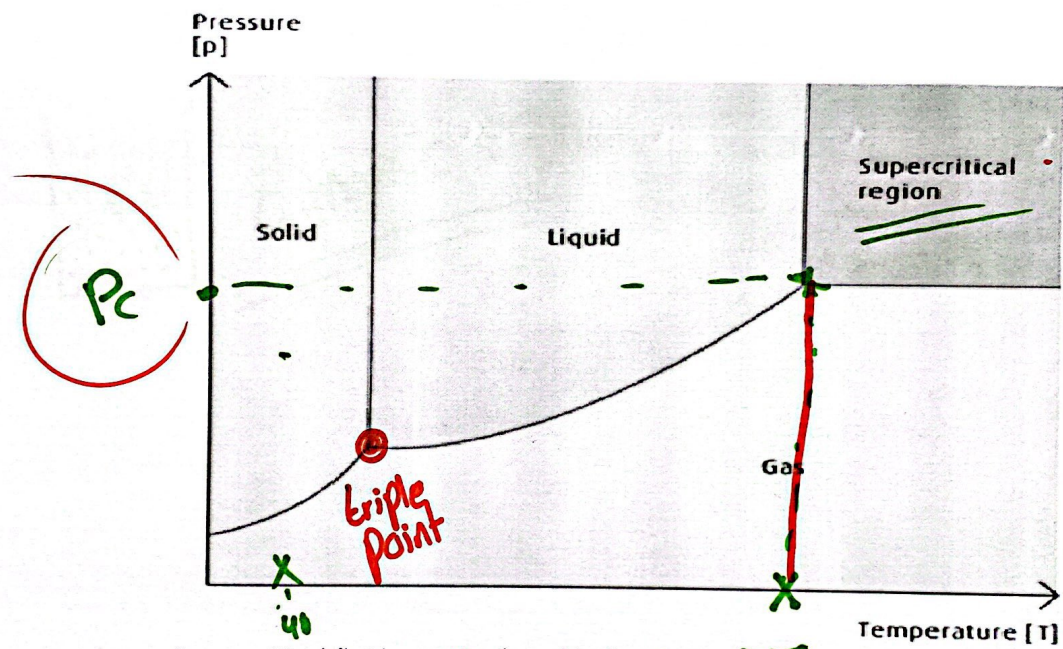


$$\Delta H_{fus} < \Delta H_{vap}$$

Heating curve

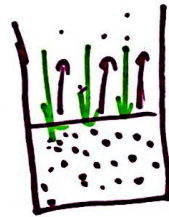


$$q = C_s * m * \Delta T$$

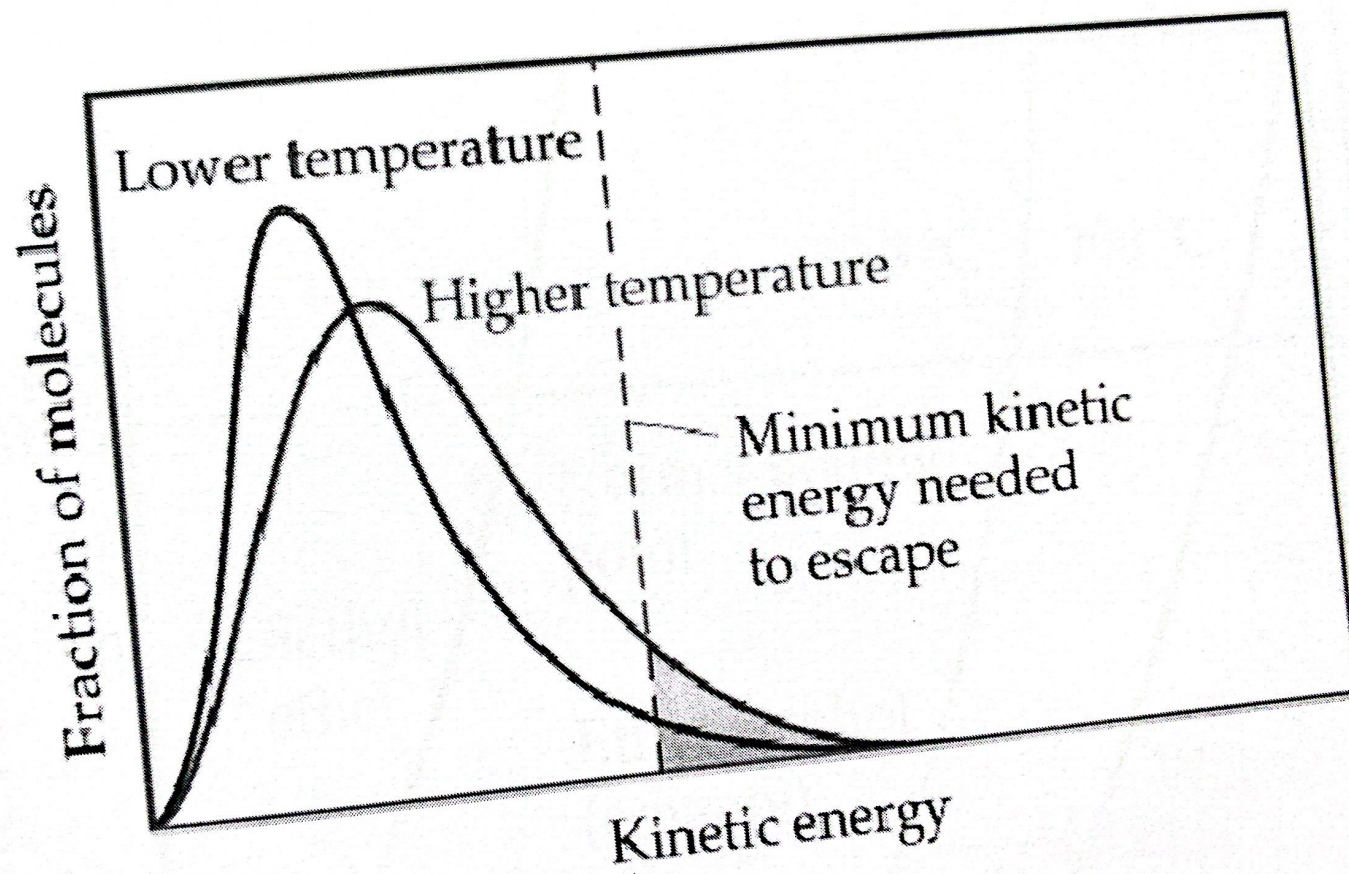


IMFT ↑
 T_c ↑
 P_c ↑

Vapor Pressure

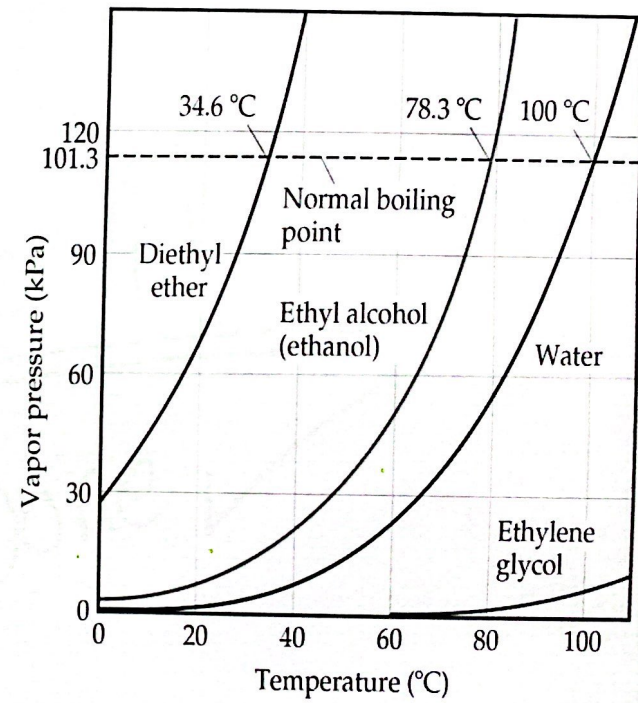


dynamic eqn



external pressure $\approx$ Vapor pressure B.P

Normal B.P
Vapor pressure = 760 torr
= 101.3 kPa



Chapter 11

Done ✓

كَيْمَاءُ اِبْرَاهِيمَ
تَشَارِكُ